Lesson plan · for the teacher

Two goals at once

Optimization · Level 4 · Lesson 1

Time
40 min

Materials
Board and chalk; Four plan cards per group (sports day), each with two scores; Graph paper (optional), to plot the plans

Learning goals

- Cross out every plan another plan beats: no worse on both goals, better on one.
- Choose among the rest with weights we agree first.
- Say what one more unit of a limit is worth.

Activity: Beaten or not?

1. Warm-up (5 min): Ask: would you rather sports day finished early, or that nobody waited long for their event? Take a quick vote and show that the class disagrees about the goal.
2. Teach (8 min): Write four plans on the board: P (3 hours, 40 pupils wait over 20 minutes), Q (3.5 hours, 25), R (3.5 hours, 30) and S (4 hours, 10). Explain when one plan beats another: no worse on both, better on one.
3. Cross out (10 min): Groups compare the plans in pairs and cross out every beaten plan. They should find that R is beaten by Q, and that P, Q and S remain.
4. Weigh (10 min): Each group agrees a weight first (how many waiting pupils is half an hour worth?), writes it down, then works out which remaining plan wins.
5. Wrap-up (7 min): Groups report their weight and their winner. Different weights, different winners: ask which weight the class would publish, and why.

Write the four plans on cards, one set per group. Graph paper helps: put hours across and waiting pupils up. A beaten plan sits straight above the plan that beats it, straight to its right, or above and to the right: R sits straight above Q (same hours, more pupils waiting).

Discussion

- Why can a beaten plan never be the right choice?
- Who should agree the weights for an exam week, and when?
- Where would one more hour or one more coin help our school most?

Exit ticket

- Name one plan you crossed out, the plan that beats it, and why.

Try it at school tomorrow
Give pupils four plans for sports day, each with two scores (how long it takes, and how many pupils wait more than 20 minutes). Groups cross out every plan another plan beats (no worse on both scores and better on one), then argue for one of the rest and say which goal they weighted more.

Practice questions · for you

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1 **In the garden plan (best profit 290), one more pupil-hour a week raises the best profit to 295 and one more bed of land to 315. The garden club can buy a bed for 20 coins or an hour of pupils' time for 10. Which is worth buying?**

◯ Neither: the plan is already the best ◯ The bed: it adds 25 for 20 coins ◯ The hour: it's cheaper

2 **Five bus routes: (total km, longest ride for any child). R1: 30 km, 55 min. R2: 34 km, 40 min. R3: 36 km, 45 min. R4: 40 km, 30 min. R5: 31 km, 60 min. Which are on the Pareto front?**

☐ R2 ☐ R3 ☐ R4 ☐ R5 ☐ R1

3 **Six exam-week plans: (name, days, pupils with two exams in a day). Finish the test inside the loop so the program keeps only the plans nobody beats. Another plan beats this one when it is no worse on both goals and better on at least one.**

```
plans = [("A", 4, 9), ("B", 5, 4),
         ("C", 5, 6), ("D", 6, 1),
         ("E", 7, 1), ("F", 6, 3)]
front = []
for name, days, clash in plans:
    beaten = False
    for other, d2, c2 in plans:
        # does other beat this plan?
        if False:
            beaten = True
    if not beaten:
        front.append(name)
print(front)
```

Write your program:

4 **Whatever positive weights you choose, the plan with the best weighted score is always on the Pareto front.**

◯ True ◯ False

5 **Exam plans A (4 days, 9 pupils clashing), B (5 days, 4) and D (6 days, 1). Score = days × weight + pupils, smallest wins. Match each weight to the winning plan.**

A day counts as 1 pupil	• • • •	A (49)
A day counts as 4 pupils	• • • •	D (7)
A day counts as 10 pupils	• • • •	B (24)

✎ **Think of a school decision where two goals pull apart. Who chose the weights last time, and were they ever said out loud?**

Check your answers

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1 **The bed: it adds 25 for 20 coins**

Solving again shows what one more unit of each limit is worth: a bed adds 25, an hour 5.

2 **R1; R2; R4**

R3 is beaten by R2 (shorter and a shorter longest ride), and R5 by R1 (30 km and 55 minutes beat 31 km and 60).

3

```
plans = [("A", 4, 9), ("B", 5, 4),
         ("C", 5, 6), ("D", 6, 1),
         ("E", 7, 1), ("F", 6, 3)]
front = []
for name, days, clash in plans:
    beaten = False
    for other, d2, c2 in plans:
        # does other beat this plan?
        ok = d2 <= days and c2 <= clash
        better = d2 < days or c2 < clash
        if ok and better:
            beaten = True
    if not beaten:
        front.append(name)
print(front)
```

A plan is dropped only when some other plan is no worse on both goals and strictly better on one.

4 **True**

If a beaten plan had the best weighted score, the plan that beats it would score at least as well on every goal and better on one, so its weighted score would be better.

5 **A day counts as 1 pupil — D (7)**
A day counts as 4 pupils — B (24)
A day counts as 10 pupils — A (49)

With weight 1: A 13, B 9, D 7. With 4: A 25, B 24, D 25.